MIL-C-60691(MU)
15 July 1969
SUPERSEDING
FA-PD-MI-2358,Rev.0
17 August 1959

MILITARY SPECIFICATION
CARTRIDGE, IMPULSE: M151 ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Impulse: M151, for use with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MS28741	- Hose Assembly, Detachable End-Fitting, Medium Pressure

DRAWINGS

Munitions Command

D10521965	- Cartridge, Impulse, M151, Assembly
C8837886	- Carton, Packing, Ammunition for Cartridge, Impulse, M151 Assembly
C8837887	- Box, Packing, Ammunition for Cartridge, Impulse, M151 Assembly

FSC 1377

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DRAWINGS (Cont'd)

IEL 10521965 - Inspection Equipment List for Cartridge, Impulse M151, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.4 Functional

3.4.1 The cartridge shall not fail to fire.

3.4.2 Ballistics.- The cartridge, when assembled, temperature conditioned, and tested in either the M9 Aircraft Canopy Remover or Cartridge Test Vehicle, Drawing No. F8648165, shall propel a 300 pound mass vertically and shall comply with the applicable requirements in Table I.

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Table I
AIRCRAFT CANOPY REMOVER, M9

Temperature Fahrenheit (F)	Velocity at Tube Separation Feet Per Second (FPS) Minimum	Peak Thrust Pounds Maximum	Ignition Delay (See 6.2) Milliseconds Maximum
-65° to -70°F	29.5	9,000	25
65° to 75°F	30.0	10,135	25
200° to 205°F	30.0	10,135	25

3.4.2.1 Initiation.- The M9 Remover shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 10 feet of aircraft hose, size no. 4, MS28471 at 70°F, plus or minus 5°F.

CARTRIDGE TEST VEHICLE, DRAWING NO. F8648165

Temperature Fahrenheit (F)	Velocity at Tube Separation Feet Per Second (FPS) Minimum	Peak Thrust Pounds Maximum	Ignition Delay (See 6.2) Milliseconds Max.
-65° to -70°F	41.0	11,310	25
65° to 75°F	46.0	12,950	25
200° to 205°F	46.0	15,855	25

3.5 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specification and the following:

3.5.1 Metal defects.- The part shall be free from cracks, splits, laminations, perforations, porosity or other metal defects.

3.5.2 Burr.- The part or assembly shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 23 plated cartridge cases and 23 plated cartridge heads for protective finish tests, 12 cartridges from ballistics for vibration tests, 50 unplated cartridge cases for hardness tests, 30 cartridges for ballistics to be tested as indicated in Table II herein: and 5 complete assemblies and 10 sets of unassembled components (including explosive components) for visual and dimensional examination.

Table II

Paragraphs				Type of Test	
Requirement	Test	Test	Sample Size	Dest	Non-Dest
3.2	4.4.1	Protective finish	23(of ea component)	X	-
3.3	4.4.2	Vibration	12(from ballistics)	-	X
Dwg.	4.4.3	Hardness	50(fr ea heat treated lot)	X	-
Dwg.	4.4.4	Airtightness	Each	-	X
3.4.2	4.4.5.1	Ballistics	30 each	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and material, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility as specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and conform to requirements as indicated in Table I. The sample for firing shall consist of the following:

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Table III

Temperature	Not Vibrated	Vibrated
-65° to -70°F	6	4
65° to 75°F	6	4
200° to 205°F	6	4

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency, an ammunition lot shall consist of a group of assembled cartridges, consisting of:

a. Components produced by a homogeneous process.

b. Propellant from not more than one lot.

c. Primers from not more than one lot.

d. Igniter from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class the criteria in Table V shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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Table IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Table V

Total Allowable Defects Per Sample Size

Sample Size for Major Defects Number of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

Sample Size for Minor Defects Number of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Primer, 50M, Dwg. D10551686 covering a detail of Drawing D10521965.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Primer foil discolored	Visual
3. Mixed primers	Visual

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.2 Igniting Disc, Drawing B8593355, covering a detail of Drawing D10521965.

CRITICAL	
None defined	
MAJOR	
101. Diameter, min.	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Thickness	Gage

4.3.2.3 Spring, Retaining, Drawing B8593610, covering a detail of Drawing D10521965.

CRITICAL	
None defined	
MAJOR	
101. Diameter outside, min.	Gage
102. Thickness	Gage
103. Metal defective	Visual

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4.3.2.3 (Cont'd)

	Defect	Method of Inspection
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual

MINOR

201.	Diameter outside, max.	Gage
202.	Width of opening when spring compressed to specified O.D.	Gage

4.3.2.4 Case, Cartridge, Drawing C10521966, covering a detail of Drawing D10521965.

CRITICAL

1.	Metal defective	Visual

MAJOR

101.	Concentricity of flange with body	Gage
102.	Straightness	Gage
103.	Diameter of flange, min.	Gage
104.	Thickness of wall at flange	Gage
105.	Thickness of base	Gage
106.	Protective finish inadequate or defective	Visual
107.	Burr or sharp corner	Visual
108.	Foreign matter	Visual

MINOR

201.	Diameter of flange, max.	Gage
202.	Diameter of body	Gage
203.	Total length	Gage
204.	Location of datum diameter	Gage
205.	Radius or chamfer missing or incorrect	Visual
206.	Surface finish improper	Visual

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4.3.2.5 Head, Cartridge, Drawing C10521967, covering a detail of Drawing D10521965.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of body with primer pocket	Gage
102. Concentricity of undercut at large end with body	Gage
103. Concentricity of inside diameter at mouth of large end with body	Gage
104. Concentricity of flash hole with primer pocket	Gage
105. Concentricity of counterbore at small end with primer pocket	Gage
106. Concentricity of diameter in front of flange with primer pocket	Gage
107. Concentricity of flange with body	Gage
108. Diameter primer pocket	Gage
109. Diameter of undercut	Gage
110. Diameter inside at mouth of large end, max.	Gage
111. Depth of primer pocket	Gage
112. Depth from large end to base of flash hole	Gage
113. Width of inside diameter behind undercut	Gage
114. Thickness at base of flash hole	Gage
115. Protective finish inadequate or defective	Visual
116. Surface finish of counterbore, small end and primer pocket improper	Visual
117. Burr or sharp corner	Visual
118. Foreign matter	Visual
MINOR	
201. Diameter of body	Gage
202. Diameter inside at mouth of large end, min.	Gage
203. Diameter of flash hole	Gage
204. Diameter of counterbore at small end	Gage
205. Diameter of body in front of flange	Gage
206. Diameter of flange	Gage
207. Depth of counterbore, small end	Gage

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4.3.2.5 (Cont'd)

	Defect	Method of Inspection
208.	Depth from face to bottom of inside diameter	Gage
209.	Total length	Gage
210.	Width of undercut, large end	Gage
211.	Location of datum diameter at flange	Gage
212.	Location of datum diameter at end of flash hole	Gage
213.	Radius or chamfer missing or incorrect	Visual
214.	Surface finish improper, except as otherwise classified	Visual

4.3.2.6 Cartridge Ring, Drawing B10521968, covering a detail of Drawing D10521965.

CRITICAL

None defined

MAJOR

101.	Concentricity of inside diameter with outside diameter	Gage
102.	Diameter inside, max.	Gage
103.	Diameter outside, min.	Gage
104.	Thickness	Gage
105.	Profile of thickness	Gage
106.	Metal defective	Visual
107.	Protective finish missing or inadequate	Visual
108.	Burr	Visual
109.	Foreign matter	Visual

MINOR

201.	Diameter inside, min.	Gage
202.	Diameter outside, max.	Gage
203.	Surface finish improper	Visual

4.3.2.7 Gasket, Drawing B10521969, covering a detail of Drawing D10521965.

CRITICAL

None defined

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity inside diameter with outside diameter	Gage
102. Diameter inside, max.	Gage
103. Diameter outside, min.	Gage
104. Thickness	Gage
105. Material defective	Visual
106. Flash	Visual
107. Foreign matter	Visual
MINOR	
201. Diameter inside, min.	Gage
202. Diameter outside, max.	Gage

4.3.2.8 Cartridge, Impulse, M151, Assembly, Drawing D10521965.

CRITICAL	
1. Any part missing	X-ray & Visual
2. Improper assembly	X-ray & Visual
3. Primer not properly seated	Visual
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of flange with body	Gage
102. Location of datum diameter	Gage
103. Primer not properly sealed and crimped	Visual
MINOR	
201. Diameter of body over marking max.	Gage
202. Diameter of flange, max.	Gage
203. Length of flange	Gage
204. Total length, max.	Gage
205. Varnish missing or inadequate	Visual
206. Marking missing, incorrect or illegible	Visual

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4.3.2.9 Carton, Packing, Ammunition for Cartridge, Impulse, M151, Drawing C8837886.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damaged through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.10 Box, Packing, Ammunition for Cartridge Impulse, M151, Drawing C8837887.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual

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4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4:

4.3.3.1 Protective finish.- Twenty-three units of each component shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Vibration.- Twelve of the 30 cartridges of the initial production sample shall be subjected to vibration. The twelve cartridges shall then be x-rayed. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot. Any change in the process of a lot of primers in subsequent production shall be reason to select an additional sample of 12 cartridges for vibration tests.

4.3.3.3 Hardness.- Fifty units from each heat treated lot of components shall be tested. Failure of any unit to comply with the requirements of the applicable drawing shall be cause for rejection of the lot of cartridges.

4.3.3.4 Airtightness.- Each cartridge shall be subjected to the specified pressure differential. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.5 Functional.- The following test shall be performed by the government:

4.3.3.5.1 Ballistics.- Thirty assembled cartridges from each production lot shall be temperature conditioned and functionally tested, as prescribed in Table VI. Failure of any cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot.

Table VI

Temperature	Sample Size
-65°F to -70°F	10
65°F to 75°F	10
200°F to 205°F	10

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL 10521965.

4.4 Test methods and procedures.

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4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case and cartridge head shall be sectioned longitudinally to better expose the significant surface to the salt spray.

4.4.2 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Hardness.- A hardness testing machine shall be used for determining the hardness values specified on the applicable drawing.

4.4.4 Airtightness.- The method for determining airtightness shall be by use of equipment listed on IEL 10521965.

4.4.5 Functional.

4.4.5.1 Ballistics.- The thirty cartridges shall be loaded into the M9 Canopy Remover or test vehicle Dwg. No. F8648165, divided into three equal groups and temperature conditioned in accordance with Table VI. The cartridge shall be tested within 10 minutes after removal from the conditioning chamber. When this time is exceeded, the remover shall be reconditioned in accordance with 4.4.5.1.1.

4.4.5.1.1 Temperature conditioning.- A reasonably like facsimile of the M9 Remover or test vehicle, Dwg. No. F8648165, loaded with a dummy M151 cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8837886 and C8837887.

5.2 Data card.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement document should specify the title, number and date of this specification.

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6.2 Ignition dealy is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Supersession data.- This specification includes the requirements of purchase description FA-PD-MI-2358 dated 17 August 1959.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A189

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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